

PUTHIN KUMAR REDDY THALLAPALLI

Network Engineer

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SUMMARY

- Network Engineer with 3+ years of experience in designing, implementing, and optimizing enterprise and cloud networks.
- Proficient in routing and switching using Cisco (ISR, ASR, Nexus), Juniper (MX, SRX), Arista, and Meraki.
- Skilled in firewall security and compliance, including Palo Alto, FortiGate, Cisco ASA, NGFW, and IDPS.
- Strong background in cloud networking with AWS (VPC, Transit Gateway, Direct Connect) and Azure (VNet, ExpressRoute).
- Expertise in network automation using Python, Ansible, Terraform, Netmiko, and Bash/Shell Scripting to streamline deployments.
- Hands-on experience in VPN (IPSec, SSL), SD-WAN, MPLS, VLANs, and QoS to enhance network performance and security.
- Implemented high-availability architectures, ensuring network reliability, scalability, and business continuity.
- Experienced in monitoring and troubleshooting using Wireshark, SolarWinds, SNMP, and Syslog for proactive network management.
- Ensured compliance with HIPAA, PCI DSS, and SOX by conducting security audits and optimizing access controls.
- Strong problem-solving, teamwork, and leadership skills, with a focus on improving network security, resilience, and efficiency.

SKILLS

- **Programming Languages:** Java, Python, C++, JavaScript, TypeScript, SQL, HTML, CSS
- **Networking Protocols:** TCP/IP, BGP, OSPF, EIGRP, IS-IS, RIP, VLANs, MPLS, SD-WAN, VPN (IPSec, SSL)
- **Routing & Switching:** Cisco (ISR, ASR, Nexus), Juniper (MX, SRX), Arista, Meraki
- **Security & Firewalls:** Palo Alto, FortiGate, Cisco ASA, NGFW, IDPS, RBAC, Encryption, Penetration Testing, PCI DSS, SOX Compliance
- **Cloud Networking:** AWS (VPC, Transit Gateway, Direct Connect), Azure (VNet, ExpressRoute), Hybrid Cloud Integration
- **Network Automation:** Python, Ansible, Terraform, Netmiko, Bash/Shell Scripting, PowerShell, Perl
- **Monitoring & Troubleshooting:** Wireshark, SolarWinds, SNMP, Syslog, Packet Tracer
- **Hardware & Infrastructure:** Cisco Routers & Switches, Palo Alto Firewalls, Aruba Wireless, Cisco WLC
- **Operating Systems:** Linux (Ubuntu, CentOS), Windows Server
- **Version Control & CI/CD:** Git, GitHub, Jenkins
- **Other Skills:** VTP, DTP, DHCP, DNS, Network Automation, Load Balancing
- **Soft Skills:** Problem-Solving, Communication, Teamwork, Leadership, Time Management, Adaptability, Attention to Detail, Customer-Oriented

WORK EXPERIENCE

Molina Healthcare

August 2024 – Present

Network Engineer

- Integrated AWS Transit Gateway and Azure ExpressRoute, reducing inter-cloud latency by 30% and enhancing secure data flow.
- Designed and implemented hybrid cloud connectivity with AWS Direct Connect and Azure VNet Peering, ensuring high availability and seamless workload distribution.
- Automated network configurations using Terraform and Ansible, reducing manual deployment efforts and configuration errors across multi-cloud environments.
- Optimized BGP route policies and traffic engineering to enhance network resilience and load balancing between on-premise and cloud infrastructures.
- Conducted security assessments and implemented RBAC, encryption, and compliance measures, ensuring adherence to HIPAA and PCI DSS regulations.

Molina Healthcare

January 2024 – March 2024

Network Engineer – Internship

- Analysed and refined Palo Alto and Cisco ASA firewall rules, reducing unnecessary rules by 40% and improving network security posture.
- Implemented VPN access controls with RBAC and MFA, enhancing authentication security and mitigating unauthorized access risks.
- Conducted firewall performance assessments, identifying and resolving bottlenecks for improved packet processing efficiency.

Infosys BPM

November 2020 – July 2022

Network Engineer

- Modernized legacy infrastructure with Cisco Nexus switches and Juniper MX routers to enhance network scalability and reliability.
- Designed and deployed VLAN segmentation strategies to optimize traffic flow and improve data security across multiple retail locations.
- Migrated critical network components to high-performance switches, reducing downtime and improving operational efficiency.
- Configured and optimized BGP and OSPF routing protocols to ensure seamless connectivity and load balancing across distributed retail sites.
- Implemented QoS policies to prioritize real-time transactions and improve payment processing speeds, reducing latency by 25% during peak hours.

- Upgraded firewall security configurations with advanced threat detection and response mechanisms, strengthening network defences.
- Deployed automated monitoring tools using SNMP and SolarWinds, enabling proactive issue detection and reducing troubleshooting time.
- Conducted rigorous performance testing and failover simulations to validate network redundancy and ensure business continuity.
- Provided technical training and documentation to internal teams, enhancing operational knowledge and streamlining future network maintenance.

Trigent

January 2020 – October 2020

Network Engineer

- Deployed Cisco ISE and 802.1X authentication to enforce endpoint security and enhance network compliance.
- Configured and managed role-based access policies (RBAC) to restrict unauthorized device connections and improve network security posture.
- Integrated Network Access Control (NAC) with Active Directory and identity management solutions to streamline user authentication processes.
- Optimized VLAN assignments and dynamic segmentation, reducing unauthorized access incidents by 40% across corporate and remote networks.
- Deployed real-time monitoring and alerting mechanisms for unauthorized access attempts, improving incident response efficiency.
- Collaborated with security and IT teams to ensure NAC policies aligned with business and regulatory compliance requirements, strengthening overall infrastructure security.

PROJECTS

Secure Payment Gateway Integration

- Developed and integrated a PCI DSS-compliant payment processing system using Stripe API, Spring Security, and JWT authentication.
- Implemented fraud detection algorithms using machine learning and behavioural analysis, reducing transaction fraud attempts.

Cloud Infrastructure Automation

- Automated AWS cloud provisioning using Terraform, ensuring repeatable and scalable infrastructure deployments.
- Integrated CI/CD pipelines with Jenkins and AWS Code Pipeline, reducing deployment time from hours to minutes.

EDUCATION

- **Master of Science in Computer Science**
University of Alabama at Birmingham
- **Bachelor of Technology in Computer Science**
Madanapalle Institute of Technology & Science

CERTIFICATIONS

- **CCNA**
- **CCNP**